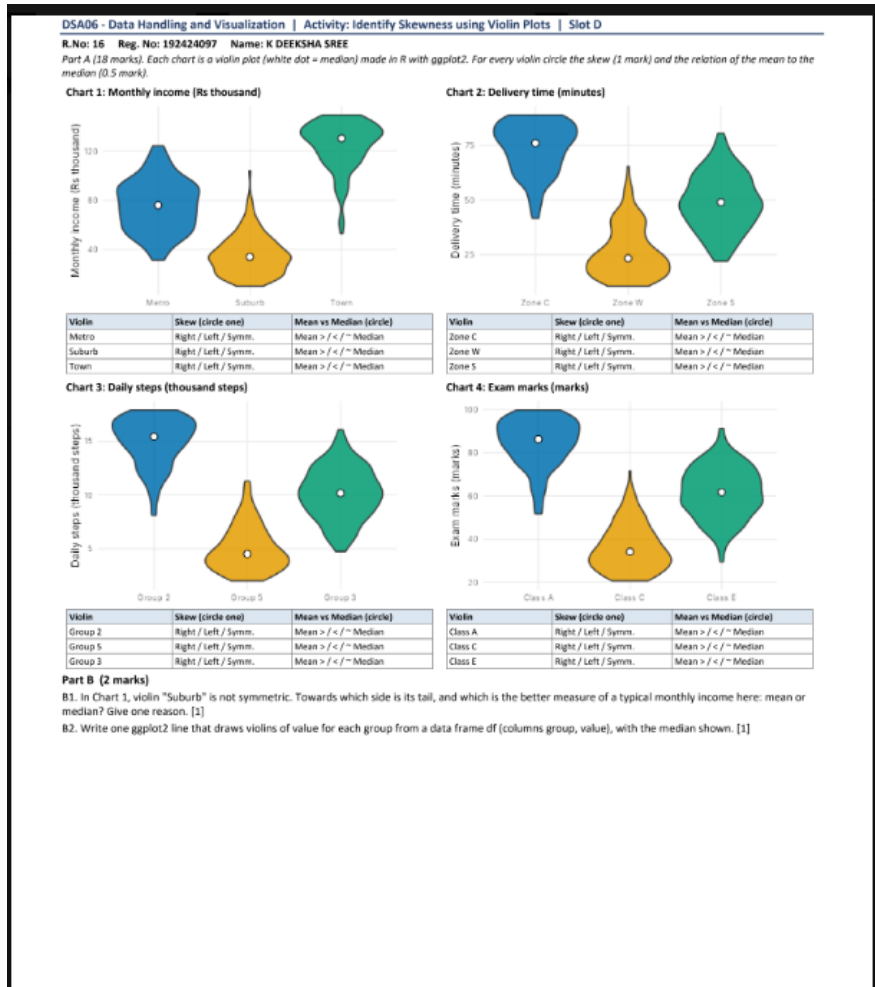


Identify Skewness using Violin Plots

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Part A – Skewness and Mean vs Median

Rule used: the white dot is the median. The long tail points toward the direction of the skew. A tail stretching upward (the violin is wide at the bottom) means right skew, so the mean is pulled above the median (Mean > Median). A tail stretching downward (wide at the top) means left skew, so Mean < Median. A balanced, mirror-like shape means symmetric, so Mean ≈ Median.

Chart 1: Monthly income (Rs thousand)

Violin	Skew (circle one)	Mean vs Median (circle)
Metro	Symmetric (circle: Symm.)	Mean ≈ Median
Suburb	Right skew (circle: Right)	Mean > Median
Town	Left skew (circle: Left)	Mean < Median

Chart 2: Delivery time

Violin	Skew (circle one)	Mean vs Median (circle)
Zone C	Left skew (circle: Left)	Mean < Median

Zone W	Right skew (circle: Right)	Mean > Median
Zone S	Symmetric (circle: Symm.)	Mean $\approx$ Median

Chart 3: Daily steps

Violin	Skew (circle one)	Mean vs Median (circle)
Group 2	Left skew (circle: Left)	Mean < Median
Group 5	Right skew (circle: Right)	Mean > Median
Group 3	Symmetric (circle: Symm.)	Mean $\approx$ Median

Chart 4: Exam marks

Violin	Skew (circle one)	Mean vs Median (circle)
Class A	Left skew (circle: Left)	Mean < Median
Class C	Right skew (circle: Right)	Mean > Median
Class E	Symmetric (circle: Symm.)	Mean $\approx$ Median

Part B

B1. In Chart 1, violin “Suburb” is not symmetric. Towards which side is its tail, and which is the better measure of a typical monthly income here: mean or median? Give one reason.

Answer: The tail of the Suburb violin points to the right / upper side (towards higher incomes), so it is right-skewed. The median is the better measure of a typical monthly income. Reason: the few very high incomes in the long tail pull the mean upward, so the mean overstates what a typical resident earns. The median is not affected by these extreme values and stays near the bulk of the data.

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown.

Answer:

```
ggplot(df, aes(x = group, y = value, fill = group)) + geom_violin() +
stat_summary(fun = median, geom = "point", colour = "white", size = 3)
```

Here geom_violin() draws one violin per group, and stat_summary(fun = median, geom = "point") adds a white dot at each group’s median, matching the charts above.